AWI–ESM3 v3.4 PI-control tuning campaign — summary

What to adopt, what reversed, and why the remaining error is a drift rather than an offset

OpenIFS 48r1 / FESOM2 / LPJ-GUESS / OASIS / XIOS, TCO95L91 / CORE3

supersedes the 2026–08–11 summary, whose headline recommendation this document reverses

2026–08–24

Abstract

The short reading of a campaign whose full record runs to 128 pages (`report.tex`). Three objectives throughout: the **global radiative imbalance**, the **Southern Ocean**, and the **NH high-latitude cold bias / boreal forest**.

Adopt `RSBLB` 2.0. Stable-boundary-layer mixing, exposed as a namelist parameter this round. Coupled, it warms the Arctic winter screen by +0.67 K, all land by +0.53 K in DJF, at *zero* net TOA cost (−0.027) and *zero* Siberian summer cost. It gives **11R the best `cmpitool` score in the campaign**, 0.7577 against the base's 0.8237, and it reverses the `tas/zg` regression that 11G introduced. It is the first lever here to move winter and summer targets in the same direction.

Retire S4 — reversing this summary's predecessor. The 2026–08–11 edition led with "Adopt S4". Measured properly, `RCL_INPPMIN` 70000 → 50000 buys nothing at the pole, authored the sub-Antarctic overcorrection that blocks every remaining Southern Ocean lever, and its headline benefit *reverses sign coupled*: Siberian JJA +1.194 K in AMIP against −0.365 K coupled. Removal is under test in 11V/11W; at 18 years it returns +3.06 W m^{-2} to the sub-Antarctic and −2.52 to the polar band, both improvements, for +0.27 W m^{-2} of energy.

The energy problem is a drift, not an offset. The PI controls start at net TOA −0.06 to −0.08 W m^{-2} and *drift* to $\sim+1.0$ over 40 years, in every arm, 1850 and 1990 alike. A TOA drifting positive while the system warms is backwards for a damped system; the cause is SH sea-ice loss of $\sim 2 \times 10^6$ km^2 over the same period. **The Southern Ocean does not respond to the imbalance, it generates it.**

The remaining land error is above the screen. At 60–90N in DJF the air-side inversion bias is +2.81 K while the skin-side bias is +0.11. Three surface levers scored null because they were adjusting a part of the column carrying no error. Of the −2.23 K all-land DJF bias, ∼1.0 K is excess inversion and ∼1.2 K is a deep tropospheric cold bias no mixing lever can reach.

1 Scorecard against the three objectives

objective	at 11E	at 11Q/11R	target	status
Global radiative imbalance	+0.778	+0.877	|<0.3|	**Not met, and reframed.** PI arms *start* at −0.06 and drift to $\sim+1.0$ over 40 yr. It is a drift, and no lever addresses it
Southern Ocean polar DJF SW CRE	+24.24	+13.36	0	45 % closed. But 60–45S went +4.20 → −3.70, i.e. past zero, and that overcorrection now blocks the remaining levers
NH cold bias Siberia JJA	−2.532	−1.532	0	A full kelvin gained, +0.767 of it from `RSBLB`. Best in the campaign
60–90N DJF	−6.149	−4.913	0	+1.24 K
all land ANN	−2.509	−1.854	0	26 % closed
Boreal forest	lost	lost	recovered	**Not met.** BNS is outcompeted by C3 grass, not gate-limited, so warming alone does not restore it

All at 1850 forcing over 1360–1389, so the comparison is internally clean; absolute values against a present-day ERA5 carry a forcing offset of order a kelvin that is not model error.

2 Where the model stands

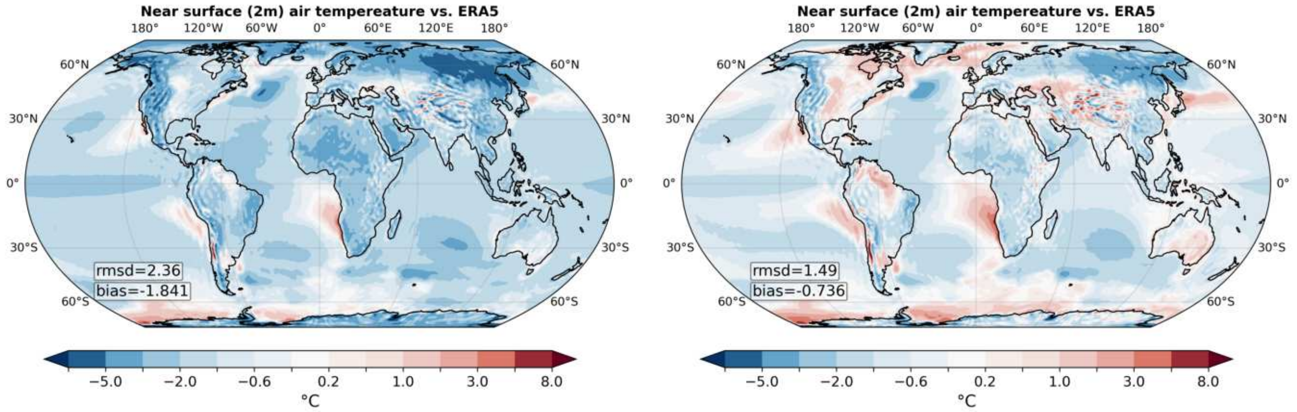

Figure 1: Near-surface temperature against ERA5: **left** 11E, the campaign base; **right** 11R, the current best. The Siberian and Arctic blues retreat and the global cold bias roughly halves. Recomputed on a consistent window and reference, the global annual bias runs −1.698 (11E) → −1.619 (11N) → −1.369 (11Q) at 1850 forcing, and −1.011 (11P) → −0.778 (11R) at 1990 — monotone improvement in both families. **Two cautions.** 11E is 1850-forced and 11R is 1990-forced, so this pair mixes tuning with a forcing change; the clean same-forcing steps are the two sequences just quoted. And the `rmsd`/`bias` annotations printed on the panels come from reval's own window and regridding and do not match the recomputed values — for 11N they even reverse the ordering against 11E, which is unexplained (§19). Note the residual warm ring around Antarctica in both panels: that is the Southern Ocean problem of §8, and it is the one feature the campaign has not moved.

Best coupled arm: **11R** (1990 forcing) and **11Q** (1850), = LX4 stack + `RSBLB` 2.0. Against 11E, at 1850 forcing so the comparison is clean:

1360–1389	11E base	11N LX4	**11Q**	11Q−11N	11Q−11E
DJF T2m 60–90N	−6.149	−5.528	**−4.913**	+0.614	+1.236
DJF T2m all land	−3.351	−3.213	**−2.647**	+0.566	+0.704
ANN T2m all land	−2.509	−2.370	**−1.854**	+0.516	+0.655
JJA T2m Siberia	−2.532	−2.299	**−1.532**	+0.767	+1.000
DJF inversion 60–90N	+3.271	+3.109	**+2.428**	−0.681	−0.843
net TOA	+0.778	+0.878	+0.877	−0.001	+0.099
NH ice Mar [10^6km^2]	14.142	13.862	13.697	−0.165	−0.445

The annual land bias is down 26 % from base and Siberian JJA has improved by a full kelvin, at unchanged net TOA. **Caveat:** the 11E→11N step spans an `lpj_guess` binary change (§18), so part of that column is not attributable to LX4.

3 The campaign in one page

Eight weeks, ∼60 atmosphere-only lever arms and ∼35 coupled arms, against three objectives that turned out to be more coupled to each other than expected.

rounds	phase	what it established
09–13	ocean/ice mixing, first levers	Aggressive coupling and KPP settings cut the imbalance hardest but cool the planet past ERA5; the cooling "leaks well beyond where an ocean-mixing parameter has any business acting". Milder settings retained
14–18	land surface, snow, aerosol	G4 adopted (`RVRSMIN` on needleleaf + tundra). The 4-year window was resolving nothing in the boreal: B8 read +0.502 at 4 yr and −0.038 at 44. Detection thresholds computed *first* from that round on
19–23	snow-cover depletion	Melt-out 13 days late and June cover twice observed; `ECE_SNOW_SCF`= 3 fitted to observations and adopted (P3). Exposed the Siberian winter cold bias the excessive cover had been masking
24–28	marine CCN, tropical longwave	DMS wired end-to-end and priced; the SO lever works and the tropical longwave gate stops it. `RSNOWLIN2` found: the only lever in ∼52 arms that moves tropical LW
29–30	land repair, SO cloud coupled	84 % of the coupled SO error is inherited from AMIP. Overlap transfers but costs 1.000 K of Siberian JJA. Ocean drift priced; the "structural leak" was omitted snow enthalpy
31	**the winter land bias relocated**	Resolved by height: the error is above the screen, not at it. Retires the surface-side lever class and explains three nulls
32	**`RSBLB` adopted**	Stable-BL mixing exposed and scored. Best `cmpitool` in the campaign; winter and summer move together for the first time
33	**the Southern Ocean reframed**	It generates the imbalance rather than responding to it. Two mechanisms, seasonally separated. S4 indicted

4 The coupled series

All branch from the same 1350 state, so differences are paired.

arm	content	outcome
11E	campaign base	the reference
11G	11E + S4	adopted base for rounds 29–30; now indicted (§7)
11I	11G + v2soil (LPJG repair)	the repair is a trade: Siberian DJF soil +0.308 K but JJA against ERA5 worse than 11E
11J	11I + Raupach z_0	null; one binary, run-time gate
11L/11M	11G + overlap 0.35 / 0.10	first SO lever to transfer (0.57–1.01×); both rejected on Siberia and TOA
11N/11P	LX4 stack, 1850 / 1990	largest single winter improvement before this round: 60–90N DJF −6.35 → −4.49
11Q/11R	+ `RSBLB` 2.0, 1850 / 1990	**current best.** 11R scores CMPI 0.7577
11V/11W	−S4, 1990 / 1850	*running*; decides S4

Read 11N→11P as a change of comparison basis, not a gain. 11N is 1850 forcing read against a present-day ERA5; 11P is period-matched. Only 11P/11R should be quoted as biases against observations.

5 The boreal forest: the founding problem, and where it stands

The campaign began because the coupled model loses the Siberian boreal forest. That is now partly understood and only partly fixed.

It is not a climate gate. BNS (larch) is permitted everywhere in Siberia and is *outcompeted by C3 grass* — established 2026-07-14 and re-derived at least five times since. Establishment thresholds, GDD5 and the climate gates are not what excludes it, so no amount of warming the gate will restore it.

Two mechanisms, both measured. A summer route — over-transpiration → moist boundary layer → JJA cloud +8.5 pp → $-7.0\,\mathrm{W\,m^{-2}}$ surface SW — and a winter route through roughness: when LPJ-GUESS loses the forest, thermal roughness collapses by three to four orders of magnitude (forest $z_{0h} = 2.0$ m against tundra 3.4×10^{-4}), the surface decouples and the inversion deepens by +2.1 K.

What worked and what did not. G4 (`RVRSMIN`) took ∼0.95 of the boreal budget and saturates: F5, combining four land levers, gives +0.746 against F4 alone at +0.749. Raupach roughness cannot bootstrap the canopy it needs. The current state is Siberian JJA −1.53 K against ERA5 in 11Q, improved a full kelvin from the base, with `RSBLB` contributing +0.767 of it — the first winter lever that also helped summer.

6 Adopt `RSBLB`: the winter land lever

Round 31 resolved the DJF land cold bias by height against ERA5. The bias decays to nothing by 850 hPa, so it is surface-confined and a redistribution lever is the right class — but the *skin is already right*:

11P, DJF	T_{2m}	T_{1000}	T_{925}	T_{850}	lowinv bias	sfcinv bias
60–90N	−3.63	−2.10	−0.88	−0.42	**+2.81**	**+0.11**
ALL LAND	−2.23	−1.43	−1.25	−1.22	+0.86	−0.36

That retires the surface-side levers as a class and explains three measured nulls — `ECE_LAMSK_SN`, Raupach z_0, F1 `RVZOH` — as the wrong layer rather than bad luck. The error sits between the screen and 925 hPa, which is the stable branch of `VDFEXCU`, whose constants were hardcoded until this round.

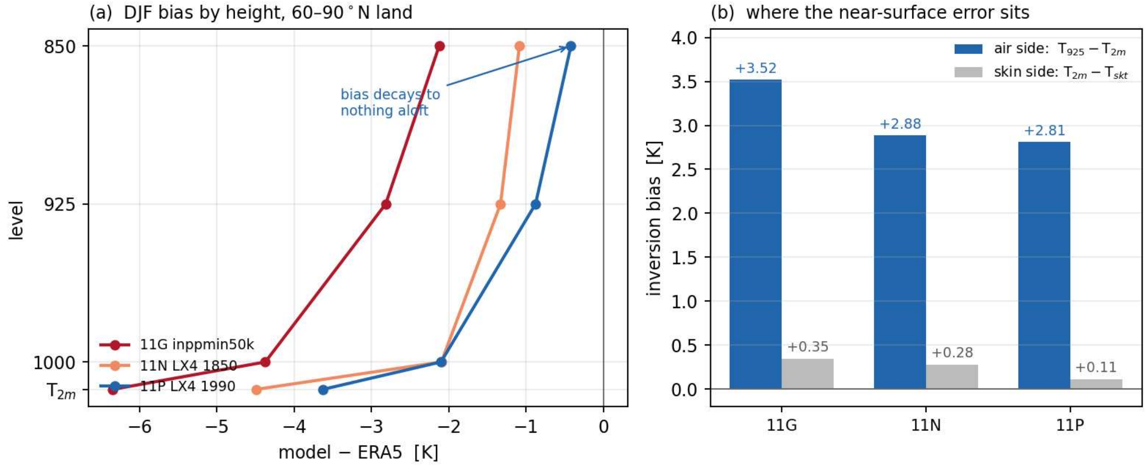

Figure 2: Why three surface levers scored null. **(a)** The DJF land cold bias over 60–90°N decays from −3.63 K at the screen to −0.42 at 850 hPa, so it is surface-confined and a redistribution lever is the right class. **(b)** But the skin-side bias is +0.11 K against an air-side +2.81: the skin is already right, and the entire excess inversion sits between the screen and 925 hPa. Skin conductivity, Raupach z_0 and F1 `RVZOH` were all adjusting a part of the column that carries no error.

Coupled, the lever performs better than AMIP predicted, because AMIP prescribes sea ice and suppresses the amplifying feedback:

1360–1389	**11Q**−11N	**11R**−11P	AMIP SB2
DJF inversion 60–90N	−0.680 *	−0.809 *	−0.713 *
DJF T2m 60–90N	**+0.614** *	**+0.674** *	+0.285
DJF T925 60–90N	−0.114	−0.020	−0.428 *
global net TOA	**−0.001**	**−0.027**	+0.175 *
Siberia JJA	+0.767 *	+0.045	+0.000

Three things AMIP could not settle all came out favourably: the screen follows (2.2–2.4× the AMIP response), the compensating cooling aloft disappears, and the +0.175 W m^{-2} TOA cost vanishes — the tropical shortwave gain is absorbed as ocean warming and cancelled by the longwave response. `cmpitool` on 11R scores **0.7577**, the best in the campaign, with `tas` −0.109 and `zg` −0.088 against 11P and `nino34` the most improved region. The cost lands on `thetao` (+0.018), i.e. ocean heat.

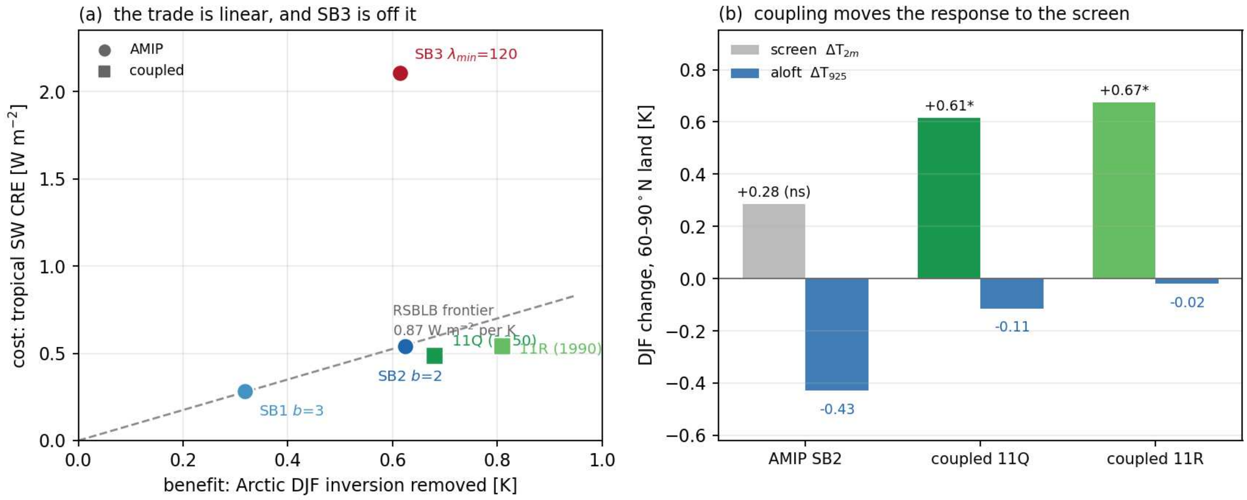

Figure 3: (a) What each arm pays (tropical SW CRE) against what it buys (Arctic DJF inversion removed). SB1 and SB2 lie on one line through the origin at $0.87\,\mathrm{W\,m^{-2}}$ per K, so the trade is *linear in dose* and cannot be dodged by tuning gentler; the mixing-length route sits far off it at 3.43. The coupled arms land on the same frontier. **(b)** Why the coupled pair was necessary: in AMIP the inversion weakens mostly by cooling the air aloft and the screen response is not significant, whereas coupled — sea ice free to respond — it is essentially all screen.

The one caveat. The DJF Arctic screen gain carries a fitted trend of −0.456 K/decade in 11Q and −0.210 in 11R over the 30 years, so the winter component may be decaying. The *annual* land and global warming does not decay.

7 Retire S4: the reversal

The predecessor to this document recommended adopting S4 on the strength of an AMIP result. Three measurements since say otherwise.

It buys nothing at the pole. Ocean-masked, 30 yr paired, 11G−11E: SW CRE at 90–60S DJF is +0.77 and *not significant*, against −2.34 * at 60–45S. A ratio of −0.33: purely equatorward.

It authored the blocking constraint. That −2.34 is the largest single contribution to the sub-Antarctic overcorrection, now $-3.70\,\mathrm{W\,m^{-2}}$ DJF, which is what disqualifies DMS, `RCL_KK_CLOUD_NUM_SEA` and every other SH-wide brightening lever — while the polar deficit stands at +13.36.

Its headline benefit reverses coupled.

	AMIP S4−control (44 yr)	coupled 11G−11E (30 yr)
Siberia JJA T2m	**+1.194** *	**−0.365** *
Arctic land JJA	+0.784 *	−0.262 *
all land ANN	+0.160 *	−0.372 *
global ANN T2m	+0.070 *	**−0.320** *

Every temperature metric flips sign. A 0.32 K global cooling is a straight cost in a model already ∼0.7 K too cold. **Removal is under test** in 11V (1990) and 11W (1850); at 18 years 11V−11R gives sub-Antarctic +3.056 *, polar −2.523 *, Siberia +0.285 (ns, right sign), net TOA +0.265 *. Both Southern Ocean bands improve. The falsifier that would keep S4 — Siberian JJA going backwards — has not fired at any length.

8 The Southern Ocean has two problems, and it generates the imbalance

Every arm drifts, including the 1850 controls: +0.8 K of SH surface air and $-2\times10^{6}\,\mathrm{km}^2$ of September ice per 40 years, with the 1990 forcing adding only +0.2 K on top. At the branch point the SH T2m bias was −0.06 to −0.65 K, i.e. the air was *not* warm — so the ocean was not warmed by the atmosphere. Both warm together, and net TOA drifts from ~-0.07 to $\sim+1.0$ over the same 40 years.

Ocean-masked, the two Southern Ocean bands are the largest positive contributors to the global imbalance, +0.32 and +0.36 W m^{-2}, against a tropical −1.09. The global figure is a small residual of large regional cancellations, so "close the TOA imbalance and the Southern Ocean follows" is circular.

Two mechanisms, seasonally separated. In DJF the sea-ice zone absorbs +7.99 W m^{-2} too much shortwave at the surface, from cloud that is present but optically thin. In MAM the anomaly is +4.48 W m^{-2} of *longwave* with almost no shortwave — autumn mixed-layer deepening entrains a +0.9 K warm anomaly sitting at exactly 100 m, which is why March ice never regrows (model 1.30 against 3.10 observed). The summer heat is stored and re-emerges in autumn; suppressing ocean mixing makes the 100 m layer *worse*, so it is fed from above, not below.

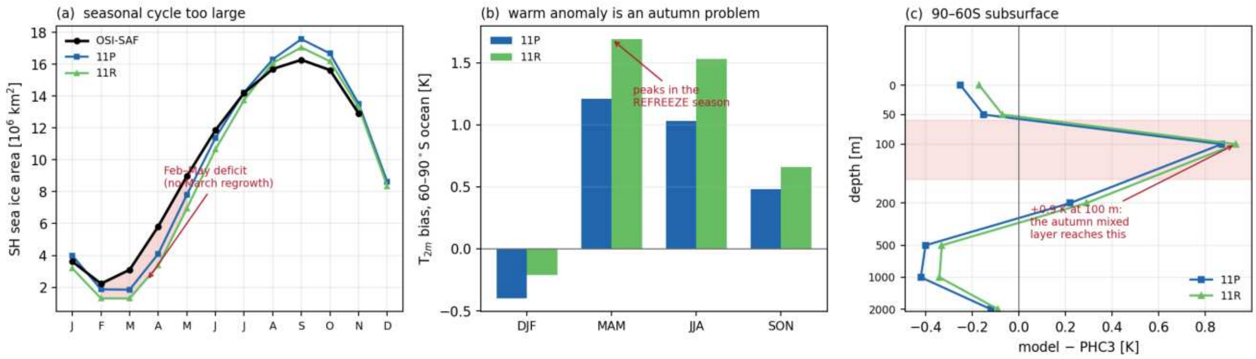

Figure 4: Two Southern Ocean problems in two different seasons. **(a)** The ice seasonal cycle is too large: September is $+0.78 \times 10^6$ km^2 in area against OSI-SAF while February–May is far too low and the March regrowth is absent entirely (model 1.30 against 3.10). **(b)** The T2m warm anomaly peaks in *MAM*, the refreeze season, which no shortwave mechanism explains. **(c)** A +0.9 K anomaly at exactly 100 m, under a neutral surface and above *cold* water at 500–1000 m — the depth the autumn mixed layer reaches. Summer heat is stored and re-emerges to block the refreeze.

The cloud error is half amount and half opacity, and the hemispheres need *opposite* corrections: Antarctic summer SW CRE is +13.36 (reflects too little) while Arctic summer is −18.22 (reflects far too much). No lever tested is selective enough to treat them separately at a useful dose.

9 The Southern Ocean cloud error in detail

All ocean-masked. Band averages fold the Antarctic ice sheet into the polar band and halve the apparent signal, which is why earlier numbers in the report read smaller.

SW CRE bias vs CERES `clr_t`	90–60S DJF	90–60S ANN	60–45S DJF	60–45S ANN
11E base	+24.24	+11.46	+4.20	+4.31
11G +S4	+21.66	+11.01	+1.75	+2.21
11N LX4	+16.50	+8.44	−0.36	+0.85
11P LX4 1990	+16.91	+8.48	−2.91	−0.29
11Q +RSBLB	+18.07	+8.96	−2.17	−0.04
11R +RSBLB 1990	**+13.36**	**+7.26**	**−3.70**	**−0.93**

A meridional gradient in lever effectiveness, not an untouched region. The polar band has been more than halved, +24.24 → +13.36. But the same levers drove 60–45S from +4.20 through zero to −3.70 — overcorrected — while the pole still leaks. Pushing any of them harder buys the last of the pole by driving the sub-Antarctic further negative, and 60–45S sits inside the campaign's own 45–65S SO metric, which is part of why this looked solved.

Half amount, half opacity. DJF, SW CRE per unit cloud cover isolates the two:

band		CRE	cover	CRE/cover
60–90S	CERES	−105.98	0.916	−115.7
60–90S	11R	−92.62	0.858	**−108.0**
60–45S	CERES	−120.75	0.912	−132.5
60–45S	11R	−116.34	0.851	**−136.7**

At the pole, cover is 5.8 pp low *and* each cloud is 6.7 % too dim, about 6.5 W m^{-2} each of the 13.36. In the sub-Antarctic there are 6.1 pp too few clouds each 3 % too *bright* — they compensate, which is how the net CRE ended up overcorrected. **Every campaign lever raises opacity**, which is why the two bands diverged.

Cloud phase flips sign at 60S. Low-level liquid fraction against CALIPSO-GOCCP: 45–65S model 0.873 vs 0.858 (too liquid, +0.015); **60–90S** 0.731 **vs** 0.759 (too icy, −0.028); Siberia 0.813 vs 0.818. Icier cloud precipitates out and carries less liquid water, which is how a cloud is present and optically thin. The INP family was retired on a phase measurement taken in the one band where the model is too liquid, and that conclusion does not extend poleward.

Where the deficit sits. Split by ice cover, DJF: the pack (1.2×10^6 km^2) is nearly unbiased at +1.47; the deficit lives in the marginal ice zone (10.0×10^6 km^2, +14.49) and the open water south of 60S (9.3×10^6 km^2, +15.86). Any candidate must act on sunlit open water beside ice, which rules out anything keyed to ice cover.

10 Sea ice against observations

SH sea ice area against OSI-SAF (1989–2014; the record contains no December):

10^6 km^2	Feb	Mar	Apr	Jun	Sep	Nov
OSI-SAF area	2.23	**3.10**	5.82	11.86	**16.27**	12.91
11R area	1.31	**1.30**	3.37	10.66	**17.05**	13.30
OSI-SAF extent	4.14	5.61	8.62	15.36	**20.23**	17.63
11R extent	2.99	—	—	—	**22.24**	—

The seasonal cycle is too large: too much ice at the September maximum (+0.78 area, +2.01 extent) and far too little through February–May, with the March regrowth absent entirely. The ice is *not* too dark — compact-ice September albedo is 0.760 at $ci \geq 0.9$ and 0.778 at $ci \geq 0.95$ against a snow albedo of 0.850, inside the observed range. But **in January–March no cell anywhere exceeds 90 % concentration**, so the area-mean albedo the sun sees falls to 0.49–0.51. The pack is too diffuse: mean concentration inside the ice edge is 0.767 in September against OSI-SAF's 0.804, spread over 2×10^6 km^2 more extent. Diffuse ice has less volume per unit area to lose and melts out rather than retreating to a compact core.

11 Forcing: what is pinned, and what is not

`NCMIPFIXYR` pins GHG, ozone, composition, stratospheric aerosol, SO_4, solar TSI *and* MACv2SP. It does **not** pin LPJ-GUESS, which carries its own `fixed_CO2`, `fixed_ndep` and `fixed_LU`. Both must be set, or a 1990 arm runs present-day climate over pre-industrial CO_2 and land use.

Three traps, all hit:

- The `&NAMECECMIP6` block *accepts* `NCMIPFIXYR` and is **never read**. Liveness must be checked as `Set JYEAR` in `NODE.001_01`. 50 AMIP arms ran transient because of it.
- `SP_SETUP: IYR` reports the MACv2SP *file* that was opened, not the year the profile is evaluated at — that is `YEAR_FR`, which honours `NCMIPFIXYR`. A correct run was cancelled on this misreading and had to be resubmitted unchanged.
- `fixed_LU` 1990 raises the peat-vs-LUH3 conflict count from 73 cells at 1850 to ∼493 at 2000; arms running the default `guess` silently over-allocate those to peat.

12 What the lever catalogue now says

A full kept/discarded table is in `tuning_summary.pdf`. The load-bearing points:

- **The stable-mixing trade is linear in dose.** Tropical SW CRE paid per K of Arctic inversion removed is 0.88, 0.87, 3.43 for `RSBLB` b=3, b=2 and the mixing-length route. Constant across a factor-two dose change, so the cost cannot be dodged by tuning gentler; `RSBLB` is the efficient frontier.
- **Two verdicts were rendered against the wrong band and are reopened.** A1c (`RDEPLIQREFDEPTH`) reads −1.30 on the polar ocean band and is nearly free. **B2** (`RCLDIFF_CONVI`) is the only arm in the catalogue that passes both polar vetoes: polar −0.68 *, Arctic +0.13 (null), sub-Antarctic +0.18 (null, the helpful sign), and Siberia +0.38 K as a bonus. It attacks the amount half that nothing else reaches.

- **DMS works at the pole but is not selective.** −6.10 at 90–60S DJF against −6.55 at 60–45S, so it deepens the overcorrection that blocks it. Tropics −2.35, global TOA −1.61. Shelved, not falsified.
- **Ocean surface albedo is not the problem.** Open-water albedo tracks CERES to within 0.007 from 70S equatorward, including the elevated 0.101 against 0.104 at 70–60S where the sun is lowest. The zenith-angle dependence works.

13 Statistics: how long an arm must run

Thresholds are computed *before* scoring, and are paired ($1.96\,\sigma/\sqrt{n}$) because arms branch from a shared state. Measured noise, from a 44-year AMIP pair:

metric	σ	n=8	n=14	n=30	n=44
60–90N DJF T2m	1.377	0.954	0.721	0.493	0.407
60–90N DJF inversion	**0.611**	0.424	0.320	0.219	0.181
Siberia JJA T2m	0.772	0.535	0.404	0.276	0.228
all-land DJF T2m	0.359	0.249	0.188	0.128	0.106
global net TOA	0.196	0.136	0.103	0.070	0.058

Score the inversion, not the screen. Differencing two levels in one column cancels the synoptic variance that dominates Arctic winter T2m: σ falls from 1.377 to 0.611, which is a factor ∼5 in years-to-decide. Resolving half the AMIP inversion bias takes 12.3 years on T2m and **2.4 on the inversion**.

Two consequences the campaign learned the hard way. A 4-year window resolves nothing in the boreal — B8's +0.502 became −0.038 at 44 years. And *proving a negative costs more than proving a positive*: bounding tropical SW CRE below 0.10 W m^{-2} needs 32 years where detecting 0.50 needs 1.3.

14 The energy budget: what closes, and what does not

Snow enthalpy closes the column. An apparent "−1.0 W m^{-2} structural leak, invariant across arms" between TOA and surface was the omitted $sf \cdot \rho \cdot L_f$ term: it measures 0.962–1.073 against residuals of −0.996 to −1.088. With it, the column conserves to 0.02–0.04 W m^{-2}. The convention is surface = $ssr + str + sshf + slhf - sf\rho L_f$, and omitting it inverted a conclusion: tuning net TOA to zero *does* stop the drift.

But the imbalance is not an offset. The PI controls start at −0.06 to −0.08 and drift to $\sim +1.0$ over 40 years. Regionally, ocean-masked, the two Southern Ocean bands contribute +0.32 and +0.36 against a tropical −1.09: the global number is a residual of large cancellations, and tuning it as a scalar treats a symptom.

Accumulated fluxes divide by 3600, both for AMIP daily and coupled monthly means — verified empirically (global ASR 240.4 W m^{-2}), not assumed from the output frequency.

15 `cmpitool`: where 11R gains and where it pays

run	CMPI	11R−11P by variable	Δ
11E base	0.8237	**tas**	**−0.109**
11G	0.8446	**zg**	**−0.088**
11N LX4 1850	0.8198	zos	−0.044
11P LX4 1990	0.7709	vas	−0.038
11R +RSBLB	**0.7577**	siconc	−0.024
		mlotst, pr, rlut, clt	−0.011 to −0.004
		so, **thetao**	+0.006, **+0.018**
		ua	**+0.024**

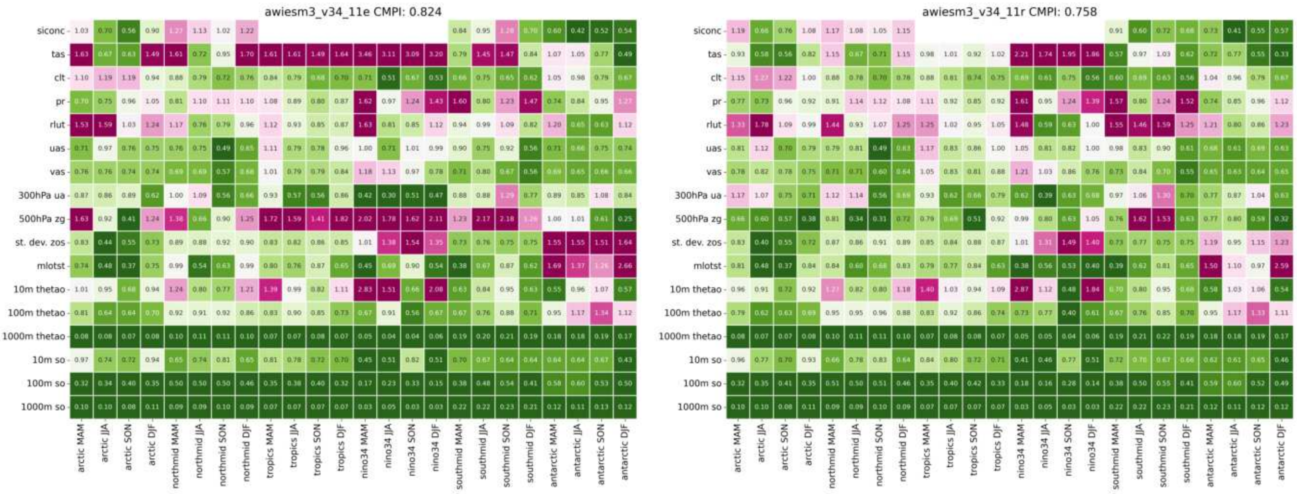

Figure 5: `cmpitool` fractional error by variable and region, **left** 11E base (CMPI 0.8237) and **right** 11R (**0.7577**, best in the campaign). Red is worse than the reference ensemble, blue better. The gain is concentrated in `tas` and `zg` — precisely the two fields where 11G had regressed against the base (+0.160, +0.149) — and `nino34` is the most improved region, contradicting the prediction that tropical low-cloud loss would damage it. The cost is `thetao`, ocean heat, which is a spin-up concern rather than a present-day skill one.

`tas` and `zg` are precisely the two fields where 11G regressed against the base (+0.160, +0.149); 11R reverses that and more. By region `nino34` is the *most* improved at −0.030, contradicting the prediction that tropical low-cloud loss would damage it. The cost is `thetao` — ocean heat, +0.051 in the tropics — which is a spin-up concern rather than a present-day skill one.

16 Levers tried and discarded

The full table with defaults is in `tuning_summary.pdf`. Struck through where the *mechanism* was falsified rather than the arm merely underperforming.

arm	parameter	why
A1a/A1b	`RCL_OVERLAPLIQICE`	Overshoots: SO CRE −6.51 but Siberian JJA −1.15 K. Coupled as 11L/11M, both rejected on Siberia and net TOA
A1c	`RDEPLIQREFDEPTH`	Reopened. "Cloud-depth selectivity dead" was rendered on the 45–65S metric; on the polar band it is −1.30 and free
A2	`RCL_KK_CLOUD_NUM_LAND`	Nothing significant anywhere
B1	`DETRPEN`	More SW yet colder; worst tropics and global RMSE
B2	`RCLDIFF_CONVI`	Reopened. Killed on a subpolar N. Atlantic RMSE; the only arm passing both polar vetoes
B3	`RCLDIFF`	Energy fails: surface flux +1.14, tropics +1.72
B4/B5/B6	`ENTSHALP`, `RCAPDCYCL`, `RLCRITSNOW`	Boreal within noise
expA/F2/F3	`RVRSMIN`, `RVLAI`, `RVCOV`	Within noise; F5 shows the land levers saturate (+0.746 against F4's +0.749)
F1	`ECE_TUNE_RVZOH`	Wrong side of the column: the skin bias is +0.11 K
B8/E1	`ECE_TUNE_RVLAMSK`	+0.502 at 4 yr was noise; −0.038 at 44
C1/C2	`RLAM`	Null — momentum mixing length, not heat
N1/N2	`RCL_INPSEA` ↓	Phase already more liquid than GOCCP at 45–65S
26	`ECE_DMS_CCN_SENS`	Works at the pole (−6.10) but hits 60–45S equally (−6.55); tropics −2.35, global TOA −1.61
SC2/SC3	`RDEPLIQREFRATE`	SC3 breaks the Arctic; rate is what destroys selectivity, not depth
06O/06T/06V	`kpp*`, `mospp`, aggressive	Improve the ocean interior but the cooling leaks to the tropics and `nino34`
W1	~~`RCLCRIT_SEA`~~	Dead parameter — never read in the KK branch
J1/J2	~~`ECE_LAMSK_SN`~~	+0.125 against a 0.41 threshold at 3.6× the default
11H/11J	~~`ifraupachz0`~~	Cannot bootstrap the canopy it needs; measured null
SB3	~~`RSBLLMIN`~~	Strips stratocumulus via the free-shear branch
D1	~~`RCAPDCYCL mode 4`~~	Mechanism falsified

17 Closed by measurement

Mechanisms falsified, not merely underperforming — these should not be retried:

claim	what killed it
~~`RCLCRIT_SEA` tunes SO cloud~~	Dead parameter: never read in the `IWARMRAIN==3` KK branch. 46 yr identical to control
~~Skin conductivity is the winter lever~~	+0.125 K at 60–90N DJF against a 0.41 threshold, at 3.6× the default. The skin-side bias is +0.11 K
~~Raupach roughness restores the boreal winter~~	Needs a canopy the model has lost ($h = 2.89$ m, $z_0 = 0.0132$ against IFS's 0.058); and measured null, DJF land–ocean gap −0.017 and −0.003
~~`RSBLLMIN` spares the tropics~~	It also sets the length in free-shear layers, so it strips stratocumulus: 4× the tropical cost of `RSBLB` for the same Arctic gain, and +1.67 W m^{-2} net TOA
~~The SO surface is too cold, so ice-albedo helps us~~	Measured with SST, which is pinned near freezing under ice. T2m reads +1.5 to +1.9 K at 70–80S
~~Tropical cloud loss will degrade `nino34`~~	`nino34` is the *most improved* region in 11R, −0.030

18 Traps that cost this campaign runs

- **Verify the staged artefact, not the source.** W1 ran 46 years identical to its control because the staged library predated the namelist exposure. `surf` parameters live in `libsurf.SP.so`, not `libarpifs.SP.so`.
- **Two `lpj_guess` binary generations exist.** `production_8c5ab467` for 11E/11G, `raupach_junebase_gated` for 11L–11R. Comparisons spanning them are confounded — including the 11E→11N step in §2 and the `RSNOWLIN2` row of the meridional decomposition. All arms are now pinned via `lpj_guess.executable`. The test built to prove the code change inert (`11HO_codeonly`) never produced output.
- **AMIP→coupled transfer has been wrong in sign**, not just magnitude — S4's Siberian JJA. Transfer ratios across the campaign span 0.19 to 1.01×.
- **Band averages fold the Antarctic ice sheet into the polar ocean**, halving the apparent signal: surface net SW bias reads −0.57 over the cap and +7.99 over ocean.
- **Sea-ice AREA and EXTENT are different numbers** and mixing them inverts the sign: September is +0.78 in area but +2.01 in extent.
- **`clr_t`, not `clr_c`**, is the model-comparable CERES clear sky. Worth 1.7 W m^{-2} at the poles.
- `NODE.001_01` is ∼2 GB with NUL bytes — `grep -a` or it silently reports nothing and looks like a failed gate.

19 Open, in priority order

1. **The drift.** +1.0 W m^{-2} and -2×10^6 km^2 of SH ice per 40 years, in every arm, addressed by no lever. It is the single largest obstacle to a multi-millennial spin-up and it is what the S4 test is ultimately about.
2. **11V/11W**, deciding S4. Read the drift rate, not the snapshot.
3. **B2 at a stronger dose** on the current base — the only untested candidate with the right meridional shape, and the only one addressing cloud amount.
4. **The Arctic contradiction**: a 13 W m^{-2} surface shortwave deficit under a surface that is +0.84 K *warm*. Unexplained, and it is the campaign's founding region.
5. **The deep ocean**, −0.33 to −0.42 K at 500–1000 m in every arm, genuinely unaddressed. The `kpplow` family is the right tool but must follow the radiative fix, not precede it.
6. **reval's printed `bias` annotations do not reproduce.** For 11N it reads −1.723 against 11E's −1.57, reversing an ordering that a consistent recomputation puts the other way (−1.619 against −1.698). Window, reference or masking — unresolved, and it affects how every reval panel should be read.
7. **reval's PHC3 ocean plots are empty for all five runs** — byte-identical 153 687-byte files. The ocean interior has never been validated by the tool.

20 The path forward

In order, with what each decides:

1. **Finish 11V/11W** (running). Decides S4. Read the *drift rate* over 40 years, not the snapshot: if removing S4 brightens the polar band by $\sim 2.5\,\mathrm{W\,m^{-2}}$ and the drift slows, the +0.27 energy penalty is repaid. If the drift is unchanged, the penalty is just a penalty.
2. **B2 at a stronger dose** against SB0, AMIP. The only untested candidate with the right meridional shape and the only one addressing cloud amount. Bracket it — its costs (tropics +0.60, TOA +0.39 at `RCLDIFF_CONVI` 10 → 25) presumably scale.
3. **Adopt `RSBLB` into the spin-up configuration** unless the Arctic screen trend continues to zero. Everything else in the stack is unchanged by it.
4. **Explain the Arctic contradiction** — $13\,\mathrm{W\,m^{-2}}$ of surface shortwave deficit under a surface +0.84 K warm. Analysis only; the runs exist.
5. **Then, and only then, the deep ocean.** −0.33 to −0.42 K at 500–1000 m in every arm. The `kpplow` family is the right tool, but applied before the radiative fix it compensates for an error that is still present — which is exactly how the 06T/06V arms ended up too cold globally.

What is not worth another arm. The mixed-phase opacity family (selectivity is dose-dependent and dies at useful doses), DMS at any dose (not polar-selective), ocean surface albedo (measured correct), and anything keyed to sea-ice cover (the deficit is beside the ice, not on it).

21 What would change the picture

- **`RSBLB` retired** if the Arctic DJF screen gain continues decaying at −0.46 K/decade and reaches zero, or if a longer coupled arm shows the sea-ice cost resolving (SH September $-1.0 \times 10^6\,\mathrm{km}^2$ in 11R is currently the one significant ice change).
- **S4 kept** if 11V/11W show Siberian JJA going backwards on removal.
- **The whole cloud route closed** if B2 at a stronger dose loses its selectivity the way `RDEPLIQREFRATE` did — selectivity has been dose-dependent every time it has been tested.
- **The drift attributed elsewhere** if 11W shows the drift rate unchanged by a $-2.5\,\mathrm{W\,m^{-2}}$ polar brightening, which would mean SH ice-albedo is not what sustains it.

22 Assets

Analysis scripts in `scripts/analysis/`: `djf_bias_vertical_structure.py` (the inversion decomposition), `coupled_sb_eval.py` (paired coupled scoring incl. sea ice), `so_subsurface_vs_phc3.py` (the 100 m anomaly; mesh-independent), `so_polar_ocean_recompute.py` (ocean-masked TOA), `polar_rescore_amip_arms.py` (the re-score that reopened A1c and B2), `stack_meridional_shape.py` (which lever is poleward-weighted), `sb_run_length.py` (how long an arm must run). Source patch: `scripts/patches/expose_sbl_mixing.py`. Build on a compute node with `scripts/build/build_oifs_compute.sh`. Parameter tables in `report/tuning_summary/tuning_summary.pdf`.